

Project Basket

AI Applications Lab (25ECP-102)

Machine Learning is an important area of Artificial Intelligence that enables systems to learn from data and make predictions or decisions. These projects are designed to give students practical exposure to real-world problems using publicly available datasets from Kaggle. The project set includes both classification and regression tasks across domains such as healthcare, finance, business, agriculture, and image processing. Through these projects, students will learn data pre-processing, model building, evaluation, and result analysis using standard machine learning techniques. This hands-on approach helps students connect theoretical concepts with practical implementation and prepares them for real-world AI applications.

DOEF-1

1. Credit Card Default Prediction (Section-1A)

- **Problem Type:** Classification
 - **Dataset:** Kaggle – Default of Credit Card Clients
 - **Objective:** Predict whether a customer will default on credit card payment
 - **Input Features:** Age, gender, education, payment history, bill amount
 - **ML Techniques:** Logistic Regression, Random Forest, XGBoost
 - **Metrics:** Accuracy, Precision, Recall, ROC-AUC
 - **Outcome:** Early identification of risky customers for financial institutions
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2. Heart Disease Classification (Section-1B)

- **Problem Type:** Classification
 - **Dataset:** Kaggle – Heart Disease Dataset
 - **Objective:** Classify patients as having heart disease or not
 - **Input Features:** Age, cholesterol, blood pressure, ECG results
 - **ML Techniques:** SVM, KNN, Decision Tree
 - **Metrics:** Accuracy, F1-score
 - **Outcome:** Supports early medical diagnosis
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3. Diabetes Prediction (Section-2A)

- **Problem Type:** Classification
- **Dataset:** Kaggle – PIMA Indians Diabetes Dataset
- **Objective:** Predict diabetes occurrence
- **Input Features:** Glucose level, BMI, insulin, age

- **ML Techniques:** Logistic Regression, Random Forest
 - **Metrics:** Accuracy, Recall
 - **Outcome:** Early detection of diabetes
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4. Titanic Survival Prediction

(Section-2B)

- **Problem Type:** Classification
 - **Dataset:** Kaggle – Titanic Dataset
 - **Objective:** Predict passenger survival
 - **Input Features:** Age, sex, ticket class, fare
 - **ML Techniques:** Naive Bayes, Random Forest
 - **Metrics:** Accuracy
 - **Outcome:** Understanding feature engineering on real data
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5. Handwritten Digit Classification (MNIST) (Section-3A)

- **Problem Type:** Classification (Image-based)
 - **Dataset:** Kaggle – MNIST
 - **Objective:** Classify digits from 0–9
 - **Input Features:** Pixel values of images
 - **ML Techniques:** CNN, ANN
 - **Metrics:** Accuracy
 - **Outcome:** Introduction to deep learning and image recognition
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6. Plant Disease Classification

(Section-3B)

- **Problem Type:** Image Classification
 - **Dataset:** Kaggle – Plant Village
 - **Objective:** Detect plant diseases from leaf images
 - **Input Features:** Leaf image pixels
 - **ML Techniques:** CNN, Transfer Learning
 - **Metrics:** Accuracy
 - **Outcome:** ML application in agriculture
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7. Dog vs Cat Image Classification

(Section-4A)

- **Problem Type:** Image Classification
- **Dataset:** Kaggle – Dogs vs Cats
- **Objective:** Classify images as dog or cat
- **ML Techniques:** CNN

- **Metrics:** Accuracy
 - **Outcome:** Practical CNN implementation
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8. Fake News Classification

(Section-4B)

- **Problem Type:** NLP Classification
 - **Dataset:** Kaggle – Fake News Dataset
 - **Objective:** Detect fake news articles
 - **Input Features:** News text
 - **ML Techniques:** TF-IDF, Logistic Regression
 - **Metrics:** Precision, Recall
 - **Outcome:** Combating misinformation
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9. Email Spam Detection

(Section-5A)

- **Problem Type:** NLP Classification
 - **Dataset:** Kaggle – Spam/Ham Dataset
 - **Objective:** Classify emails as spam or not
 - **ML Techniques:** Naive Bayes, SVM
 - **Metrics:** Accuracy
 - **Outcome:** Email filtering systems
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10. Customer Churn Prediction

(Section-5B)

- **Problem Type:** Classification
 - **Dataset:** Kaggle – Telecom Churn
 - **Objective:** Predict customer churn
 - **Input Features:** Usage, billing, contract type
 - **ML Techniques:** Random Forest
 - **Metrics:** ROC-AUC
 - **Outcome:** Business decision support
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11. Sentiment Analysis of Tweets

(Section-6A)

- **Problem Type:** NLP Classification
- **Dataset:** Kaggle – Sentiment140
- **Objective:** Classify tweet sentiments
- **ML Techniques:** TF-IDF, Logistic Regression
- **Metrics:** Accuracy
- **Outcome:** Opinion mining

12. Loan Approval Prediction

(Section-6B)

- **Problem Type:** Classification
- **Dataset:** Kaggle – Loan Prediction
- **Objective:** Predict loan approval
- **Input Features:** Income, credit history
- **ML Techniques:** Decision Tree
- **Metrics:** Accuracy
- **Outcome:** Risk assessment automation

13. Breast Cancer Classification

(Section-7A)

Problem Type: Classification

Dataset: Kaggle – Breast Cancer Wisconsin Dataset

Objective: Classify breast tumors as benign or malignant using diagnostic features.

Input Features: Cell radius, texture, perimeter, smoothness

ML Techniques: Logistic Regression, SVM, Random Forest

Metrics: Accuracy, Precision, Recall, F1-score

Outcome: Assists doctors in early and accurate cancer diagnosis.

14. Wine Quality Classification

(Section-7B)

Problem Type: Classification

Dataset: Kaggle – Wine Quality Dataset

Objective: Classify wine samples into quality categories based on chemical properties.

Input Features: Alcohol, acidity, pH, sulphates

ML Techniques: Decision Tree, Random Forest

Metrics: Accuracy

Outcome: Helps understand quality prediction in food industries.

15. Iris Flower Classification

(Section-8A)

Problem Type: Classification

Dataset: Kaggle – Iris Dataset

Objective: Classify iris flowers into different species.

Input Features: Sepal length, sepal width, petal length, petal width

ML Techniques: KNN, Logistic Regression

Metrics: Accuracy

Outcome: Introductory project for understanding supervised learning.

16. Customer Segmentation Classification (Section-8B)

Problem Type: Classification

Dataset: Kaggle – Mall Customer Dataset

Objective: Classify customers into predefined groups based on purchasing behavior.

Input Features: Age, income, spending score

ML Techniques: Decision Tree, Random Forest

Metrics: Accuracy

Outcome: Helps businesses personalize marketing strategies.

17. Bank Marketing Prediction (Section-9A)

Problem Type: Classification

Dataset: Kaggle – Bank Marketing Dataset

Objective: Predict whether a customer will subscribe to a bank term deposit.

Input Features: Age, job, balance, previous campaign outcome

ML Techniques: Logistic Regression, Random Forest

Metrics: Accuracy, ROC-AUC

Outcome: Improves marketing campaign effectiveness.

18. Employee Attrition Prediction (Section-9B)

Problem Type: Classification

Dataset: Kaggle – HR Analytics Dataset

Objective: Predict whether an employee is likely to leave the organization.

Input Features: Job role, salary, experience, work-life balance

ML Techniques: Decision Tree, Random Forest

Metrics: Accuracy, Recall

Outcome: Supports HR retention strategies.

19. Traffic Accident Severity Classification (Section-10A)

Problem Type: Classification

Dataset: Kaggle – Traffic Accident Dataset

Objective: Classify road accidents based on severity level.

Input Features: Weather, time, vehicle type, road condition

ML Techniques: Random Forest, SVM

Metrics: Accuracy

Outcome: Assists traffic safety analysis and planning.

20. Movie Genre Classification

(Section-10B)

Problem Type: NLP Classification

Dataset: Kaggle – Movie Metadata Dataset

Objective: Classify movies into genres using textual descriptions.

Input Features: Plot summaries, keywords

ML Techniques: TF-IDF, Naive Bayes

Metrics: Accuracy

Outcome: Content recommendation and media analysis.

DOEF-2

21. SMS Spam Classification (Section-1A)

Problem Type: NLP Classification
Dataset: Kaggle – SMS Spam Collection
Objective: Classify SMS messages as spam or legitimate.
Input Features: Message text
ML Techniques: Naive Bayes, Logistic Regression
Metrics: Accuracy
Outcome: Mobile communication filtering systems.

22. Student Performance Classification (Section-1B)

Problem Type: Classification
Dataset: Kaggle – Student Performance Dataset
Objective: Classify students based on academic performance.
Input Features: Attendance, study time, previous scores
ML Techniques: Decision Tree, Random Forest
Metrics: Accuracy
Outcome: Identifies students needing academic support.

23. COVID-19 Case Severity Classification (Section-2A)

Problem Type: Classification
Dataset: Kaggle – COVID Clinical Dataset
Objective: Classify COVID-19 cases by severity level.
Input Features: Oxygen level, age, comorbidities
ML Techniques: Logistic Regression, Random Forest
Metrics: Accuracy, Recall
Outcome: Supports healthcare prioritization.

24. Credit Risk Classification (Section-2B)

Problem Type: Classification
Dataset: Kaggle – Credit Risk Dataset
Objective: Classify borrowers as low-risk or high-risk.
Input Features: Credit history, income, loan amount
ML Techniques: Logistic Regression, Random Forest

Metrics: Accuracy, ROC-AUC
Outcome: Improves loan risk assessment.

25. Online Fraud Detection (Section-3A)

Problem Type: Classification
Dataset: Kaggle – Fraud Detection Dataset
Objective: Detect fraudulent financial transactions.
Input Features: Transaction amount, frequency, time
ML Techniques: Random Forest, XGBoost
Metrics: Precision, Recall
Outcome: Prevents financial fraud.

26. Product Review Polarity Classification (Section-3B)

Problem Type: NLP Classification
Dataset: Kaggle – Product Reviews Dataset
Objective: Classify product reviews as positive or negative.
Input Features: Review text
ML Techniques: TF-IDF, Logistic Regression
Metrics: Accuracy
Outcome: Customer sentiment analysis.

27. Customer Complaint Classification (Section-4A)

Problem Type: NLP Classification
Dataset: Kaggle – Consumer Complaints Dataset
Objective: Classify complaints into issue categories.
Input Features: Complaint text
ML Techniques: Naive Bayes, SVM
Metrics: Accuracy
Outcome: Improves customer service response.

28. Personality Prediction (Section-4B)

Problem Type: Classification
Dataset: Kaggle – Personality Dataset
Objective: Predict personality traits based on questionnaire data.
Input Features: Survey responses
ML Techniques: Logistic Regression, Random Forest

Metrics: Accuracy

Outcome: Behavioral analysis applications.

29. Insurance Claim Approval Prediction (**Section-5A**)

Problem Type: Classification

Dataset: Kaggle – Insurance Dataset

Objective: Predict whether an insurance claim will be approved.

Input Features: Policy type, claim amount, customer history

ML Techniques: Decision Tree, Random Forest

Metrics: Accuracy

Outcome: Automates insurance decision processes.

30. Hotel Booking Cancellation Prediction (**Section-5B**)

Problem Type: Classification

Dataset: Kaggle – Hotel Booking Dataset

Objective: Predict whether a hotel booking will be cancelled.

Input Features: Booking date, customer type, deposit

ML Techniques: Random Forest

Metrics: Accuracy

Outcome: Revenue optimization for hotels.

31. House Price Prediction (**Section-6A**)

Problem Type: Regression

Dataset: Kaggle – House Prices Dataset

Objective: Predict house prices based on property and location features.

Input Features: Area, number of rooms, location, year built

ML Techniques: Linear Regression, Random Forest Regressor

Metrics: MAE, RMSE, R^2

Outcome: Supports real estate price estimation and planning.

32. Car Price Prediction (**Section-6B**)

Problem Type: Regression

Dataset: Kaggle – Used Car Dataset

Objective: Predict the resale price of used cars.

Input Features: Brand, mileage, engine type, age of vehicle

ML Techniques: Linear Regression, Random Forest Regressor

Metrics: MAE, RMSE

Outcome: Helps buyers and sellers estimate fair car prices.

33. Medical Insurance Cost Prediction (**Section-7A**)

Problem Type: Regression

Dataset: Kaggle – Medical Insurance Dataset

Objective: Predict medical insurance charges using health-related attributes.

Input Features: Age, BMI, smoking habits, region

ML Techniques: Linear Regression, Gradient Boosting

Metrics: MAE, RMSE, R^2

Outcome: Supports insurance premium estimation.

34. Student Score Prediction (**Section-7B**)

Problem Type: Regression

Dataset: Kaggle – Student Performance Dataset

Objective: Predict student examination scores based on academic factors.

Input Features: Study hours, attendance, previous scores

ML Techniques: Linear Regression, Decision Tree Regressor

Metrics: MAE, RMSE

Outcome: Helps educators evaluate academic performance trends.

35. Salary Prediction (**Section-8A**)

Problem Type: Regression

Dataset: Kaggle – Salary Dataset

Objective: Predict employee salary based on experience and education level.

Input Features: Years of experience, qualification, job role

ML Techniques: Linear Regression, Random Forest Regressor

Metrics: MAE, RMSE, R^2

Outcome: Assists in compensation planning.

36. Flight Fare Prediction (**Section-8B**)

Problem Type: Regression

Dataset: Kaggle – Flight Fare Dataset

Objective: Predict airline ticket prices using travel details.

Input Features: Airline, journey date, source, destination

ML Techniques: Random Forest Regressor, XGBoost

Metrics: MAE, RMSE

Outcome: Helps travelers and airlines in price estimation.

37. Bike Sharing Demand Prediction (Section-9A)

Problem Type: Regression

Dataset: Kaggle – Bike Sharing Dataset

Objective: Predict the demand for bike rentals.

Input Features: Weather, season, time of day

ML Techniques: Linear Regression, Random Forest Regressor

Metrics: RMSE, R^2

Outcome: Supports urban transportation planning.

38. Energy Consumption Prediction (Section-9B)

Problem Type: Regression

Dataset: Kaggle – Energy Consumption Dataset

Objective: Predict household energy consumption patterns.

Input Features: Time, temperature, historical usage

ML Techniques: Linear Regression, Gradient Boosting

Metrics: MAE, RMSE

Outcome: Enables efficient energy management.

39. Stock Price Prediction (Section-10A)

Problem Type: Regression

Dataset: Kaggle – Stock Market Dataset

Objective: Predict stock prices using historical trading data.

Input Features: Open, close, volume, moving averages

ML Techniques: Linear Regression, LSTM

Metrics: RMSE

Outcome: Financial trend analysis and forecasting.

40. Crop Yield Prediction (Section-10B)

Problem Type: Regression

Dataset: Kaggle – Crop Yield Dataset

Objective: Predict agricultural crop yield using climate data.

Input Features: Rainfall, temperature, soil type

ML Techniques: Random Forest Regressor

Metrics: RMSE, R^2

Outcome: Supports agricultural planning and decision-making.

DOEF-3

41. Real Estate Price Prediction

Problem Type: Regression

Dataset: Kaggle – Real Estate Dataset

Objective: Predict property prices based on economic and location factors.

Input Features: Distance to city, area, age of building

ML Techniques: Linear Regression, Random Forest Regressor

Metrics: MAE, RMSE

Outcome: Real estate valuation support.

42. Sales Forecasting

Problem Type: Regression

Dataset: Kaggle – Sales Forecast Dataset

Objective: Predict future sales based on historical data.

Input Features: Past sales, promotions, seasonality

ML Techniques: Linear Regression, XGBoost

Metrics: RMSE

Outcome: Helps businesses plan inventory and demand.

43. Restaurant Revenue Prediction

Problem Type: Regression

Dataset: Kaggle – Restaurant Revenue Dataset

Objective: Predict restaurant revenue using operational data.

Input Features: Location, seating capacity, promotions

ML Techniques: Random Forest Regressor

Metrics: RMSE, R^2

Outcome: Business performance forecasting.

44. Taxi Fare Prediction

Problem Type: Regression
Dataset: Kaggle – Taxi Fare Dataset
Objective: Predict taxi fare amounts based on trip details.
Input Features: Distance, time, passenger count
ML Techniques: Linear Regression, Random Forest Regressor
Metrics: RMSE
Outcome: Fare estimation systems.

45. Gold Price Prediction

Problem Type: Regression
Dataset: Kaggle – Gold Price Dataset
Objective: Predict gold prices using historical economic data.
Input Features: Previous prices, market indicators
ML Techniques: Linear Regression, LSTM
Metrics: RMSE
Outcome: Investment trend analysis.

46. Weather Temperature Prediction

Problem Type: Regression
Dataset: Kaggle – Weather Dataset
Objective: Predict daily temperature values.
Input Features: Humidity, pressure, historical temperature
ML Techniques: Linear Regression
Metrics: MAE, RMSE
Outcome: Weather forecasting support.

47. Population Growth Prediction

Problem Type: Regression
Dataset: Kaggle – Population Dataset
Objective: Predict future population growth trends.
Input Features: Historical population data
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Urban and policy planning.

48. Water Quality Index Prediction

Problem Type: Regression

Dataset: Kaggle – Water Quality Dataset

Objective: Predict water quality index using chemical parameters.

Input Features: pH, turbidity, dissolved oxygen

ML Techniques: Random Forest Regressor

Metrics: RMSE, R^2

Outcome: Environmental monitoring.

49. Electricity Load Forecasting

Problem Type: Regression

Dataset: Kaggle – Electricity Load Dataset

Objective: Predict electricity demand based on consumption history.

Input Features: Time, load history, temperature

ML Techniques: Linear Regression, LSTM

Metrics: RMSE

Outcome: Power grid management.

50. Air Pollution Level Prediction

Problem Type: Regression

Dataset: Kaggle – Air Quality Dataset

Objective: Predict air pollution levels (AQI) using environmental factors.

Input Features: PM2.5, PM10, temperature, humidity

ML Techniques: Random Forest Regressor

Metrics: RMSE, R^2

Outcome: Air quality monitoring and control.

DOEF-4

51. Face Mask Detection

Problem Type: Image Classification

Dataset: Kaggle – Face Mask Detection Dataset

Objective: Detect whether a person is wearing a face mask or not using image data.

Input Features: Facial image pixels

ML Techniques: CNN, Transfer Learning

Metrics: Accuracy, Precision, Recall

Outcome: Supports public health monitoring and safety enforcement.

52. Credit Score Prediction

Problem Type: Classification

Dataset: Kaggle – Credit Score Dataset

Objective: Classify individuals into different credit score categories based on financial behavior.

Input Features: Income, credit history, outstanding loans

ML Techniques: Logistic Regression, Random Forest, XGBoost

Metrics: Accuracy, ROC-AUC

Outcome: Assists financial institutions in creditworthiness assessment.

53. Fake Job Posting Detection

Problem Type: NLP Classification

Dataset: Kaggle – Fake Job Postings

Objective: Identify fraudulent job advertisements.

Input Features: Job title, description, company profile

ML Techniques: TF-IDF, Logistic Regression, Random Forest

Metrics: Accuracy, Precision, Recall

Outcome: Helps job portals prevent fraud.

54. Credit Card Fraud Amount Prediction

Problem Type: Regression

Dataset: Kaggle – Credit Card Transactions

Objective: Predict financial loss due to fraudulent transactions.

Input Features: Transaction amount, time, merchant details

ML Techniques: Linear Regression, Random Forest Regressor

Metrics: MAE, RMSE

Outcome: Financial risk estimation.

55. Employee Salary Hike Prediction

Problem Type: Regression

Dataset: Kaggle – HR Analytics

Objective: Predict salary increment percentage.

Input Features: Experience, performance rating, role

ML Techniques: Linear Regression, Gradient Boosting

Metrics: RMSE, R^2

Outcome: HR decision support.

56. Disease Symptom Classification

Problem Type: Classification

Dataset: Kaggle – Disease Symptom Dataset

Objective: Classify diseases based on symptoms.

Input Features: Symptom indicators

ML Techniques: Decision Tree, Random Forest

Metrics: Accuracy

Outcome: Medical decision assistance.

57. Resume Screening Automation

Problem Type: NLP Classification

Dataset: Kaggle – Resume Dataset

Objective: Classify resumes as suitable or unsuitable.

Input Features: Resume text

ML Techniques: TF-IDF, SVM

Metrics: Accuracy

Outcome: Automated recruitment support.

58. Online Course Completion Prediction

Problem Type: Classification

Dataset: Kaggle – Online Learning Dataset

Objective: Predict whether a learner will complete a course.

Input Features: Login frequency, assignment scores

ML Techniques: Logistic Regression, Random Forest

Metrics: Accuracy

Outcome: Improves course design.

59. Student Dropout Prediction

Problem Type: Classification

Dataset: Kaggle – Education Dataset

Objective: Predict student dropout risk.

Input Features: Attendance, grades, engagement

ML Techniques: Decision Tree, Random Forest

Metrics: Recall, Accuracy

Outcome: Early academic intervention.

60. Hotel Review Rating Prediction

Problem Type: Regression

Dataset: Kaggle – Hotel Reviews

Objective: Predict hotel ratings from review text.

Input Features: Review content

ML Techniques: TF-IDF, Linear Regression

Metrics: RMSE

Outcome: Customer feedback analysis.

DOEF-5

61. Social Media Engagement Prediction

Problem Type: Regression

Dataset: Kaggle – Social Media Analytics

Objective: Predict likes or shares on posts.

Input Features: Post length, hashtags, time

ML Techniques: Random Forest Regressor

Metrics: RMSE

Outcome: Digital marketing optimization.

62. News Topic Classification

Problem Type: NLP Classification

Dataset: Kaggle – News Articles

Objective: Classify news into topic categories.

Input Features: Article text

ML Techniques: Naive Bayes, SVM

Metrics: Accuracy

Outcome: News organization systems.

63. Mobile Price Range Prediction

Problem Type: Classification

Dataset: Kaggle – Mobile Price Dataset

Objective: Classify phones into price ranges.

Input Features: RAM, battery, camera

ML Techniques: Random Forest, Decision Tree

Metrics: Accuracy

Outcome: Product pricing analysis.

64. E-commerce Product Demand Prediction

Problem Type: Regression
Dataset: Kaggle – Sales Dataset
Objective: Predict product demand.
Input Features: Past sales, promotions
ML Techniques: Linear Regression, XGBoost
Metrics: RMSE
Outcome: Inventory management.

65. YouTube Video Popularity Prediction

Problem Type: Regression
Dataset: Kaggle – YouTube Statistics
Objective: Predict video popularity.
Input Features: Views, duration, tags
ML Techniques: Random Forest Regressor
Metrics: RMSE
Outcome: Content strategy planning.

66. Disease Progression Prediction

Problem Type: Regression
Dataset: Kaggle – Medical Progression Data
Objective: Predict disease progression stage.
Input Features: Clinical test results
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Healthcare monitoring.

67. Vehicle Accident Risk Prediction

Problem Type: Classification
Dataset: Kaggle – Road Safety Data
Objective: Predict accident risk.
Input Features: Driver age, weather, speed
ML Techniques: Random Forest
Metrics: Accuracy
Outcome: Traffic safety improvement.

68. Online Advertisement Click Prediction

Problem Type: Classification
Dataset: Kaggle – Ad Click Dataset
Objective: Predict ad click behavior.
Input Features: User profile, ad content
ML Techniques: Logistic Regression
Metrics: ROC-AUC
Outcome: Digital advertising optimization.

69. Crop Disease Severity Prediction

Problem Type: Classification
Dataset: Kaggle – Crop Disease Dataset
Objective: Predict severity of crop disease.
Input Features: Leaf images
ML Techniques: CNN
Metrics: Accuracy
Outcome: Precision agriculture.

70. Food Delivery Time Prediction

Problem Type: Regression
Dataset: Kaggle – Food Delivery Dataset
Objective: Predict delivery time.
Input Features: Distance, traffic, restaurant
ML Techniques: Random Forest Regressor
Metrics: RMSE
Outcome: Logistics optimization.

71. Mental Health Risk Classification

Problem Type: Classification
Dataset: Kaggle – Mental Health Survey
Objective: Classify mental health risk levels.
Input Features: Survey responses
ML Techniques: Logistic Regression
Metrics: Accuracy
Outcome: Health awareness.

72. Smart Parking Occupancy Prediction

Problem Type: Regression
Dataset: Kaggle – Parking Dataset
Objective: Predict parking slot availability.
Input Features: Time, location, history
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Smart city planning.

73. Power Outage Prediction

Problem Type: Classification
Dataset: Kaggle – Power Grid Data
Objective: Predict power outage events.
Input Features: Weather, load
ML Techniques: Random Forest
Metrics: Accuracy
Outcome: Grid reliability.

74. Customer Lifetime Value Prediction

Problem Type: Regression
Dataset: Kaggle – Customer Analytics
Objective: Predict customer lifetime value.
Input Features: Purchase frequency, spending
ML Techniques: Gradient Boosting
Metrics: RMSE
Outcome: Business growth planning.

75. Loan Repayment Duration Prediction

Problem Type: Regression
Dataset: Kaggle – Loan Dataset
Objective: Predict repayment duration.
Input Features: Income, loan amount
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Credit planning.

76. Traffic Flow Prediction

Problem Type: Regression
Dataset: Kaggle – Traffic Flow Dataset
Objective: Predict traffic density.
Input Features: Time, location
ML Techniques: Random Forest
Metrics: RMSE
Outcome: Traffic management.

77. Disease Outbreak Prediction

Problem Type: Classification
Dataset: Kaggle – Epidemiology Data
Objective: Predict disease outbreak risk.
Input Features: Case counts, region
ML Techniques: Logistic Regression
Metrics: Accuracy
Outcome: Public health planning.

78. Insurance Premium Prediction

Problem Type: Regression
Dataset: Kaggle – Insurance Dataset
Objective: Predict insurance premiums.
Input Features: Age, health factors
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Pricing optimization.

79. Shopping Cart Abandonment Prediction

Problem Type: Classification
Dataset: Kaggle – E-commerce Dataset
Objective: Predict cart abandonment.
Input Features: Session duration, clicks
ML Techniques: Random Forest
Metrics: Accuracy
Outcome: Sales improvement.

80. Music Genre Prediction

Problem Type: Classification
Dataset: Kaggle – Music Features
Objective: Predict music genre.
Input Features: Tempo, pitch, rhythm
ML Techniques: SVM
Metrics: Accuracy
Outcome: Audio recommendation systems.

81. Student Placement Prediction

Problem Type: Classification
Dataset: Kaggle – Placement Dataset
Objective: Predict student placement outcome.
Input Features: CGPA, skills, internships
ML Techniques: Logistic Regression
Metrics: Accuracy
Outcome: Career guidance.

82. House Rent Prediction

Problem Type: Regression
Dataset: Kaggle – House Rent Dataset
Objective: Predict monthly rent.
Input Features: Location, size
ML Techniques: Random Forest Regressor
Metrics: RMSE
Outcome: Rental pricing.

83. Customer Satisfaction Prediction

Problem Type: Regression
Dataset: Kaggle – Feedback Dataset
Objective: Predict satisfaction score.
Input Features: Ratings, feedback
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Service quality improvement.

84. Road Condition Classification

Problem Type: Classification
Dataset: Kaggle – Road Images
Objective: Classify road conditions.
Input Features: Image pixels
ML Techniques: CNN
Metrics: Accuracy
Outcome: Road maintenance planning.

85. Crop Price Prediction

Problem Type: Regression
Dataset: Kaggle – Crop Market Prices
Objective: Predict crop prices.
Input Features: Market trends, season
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Farmer decision support.

86. Employee Performance Rating Prediction

Problem Type: Regression
Dataset: Kaggle – HR Dataset
Objective: Predict performance ratings.
Input Features: Experience, KPIs
ML Techniques: Random Forest
Metrics: RMSE
Outcome: Workforce evaluation.

87. Health Insurance Fraud Detection

Problem Type: Classification
Dataset: Kaggle – Insurance Claims
Objective: Detect fraudulent claims.
Input Features: Claim amount, history
ML Techniques: Random Forest
Metrics: Precision, Recall
Outcome: Fraud prevention.

88. Air Traffic Delay Prediction

Problem Type: Regression
Dataset: Kaggle – Flight Delay Data
Objective: Predict flight delays.
Input Features: Weather, schedule
ML Techniques: XGBoost
Metrics: RMSE
Outcome: Aviation planning.

89. Online Review Helpfulness Prediction

Problem Type: Classification
Dataset: Kaggle – Review Dataset
Objective: Predict review helpfulness.
Input Features: Review text, length
ML Techniques: Logistic Regression
Metrics: Accuracy
Outcome: Recommendation systems.

90. Energy Demand Peak Prediction

Problem Type: Classification
Dataset: Kaggle – Energy Data
Objective: Predict peak demand periods.
Input Features: Time, consumption
ML Techniques: Random Forest
Metrics: Accuracy
Outcome: Energy planning.

91. Customer Purchase Frequency Prediction

Problem Type: Regression
Dataset: Kaggle – Retail Dataset
Objective: Predict purchase frequency.
Input Features: Past purchases
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Marketing strategy.

92. Smart Water Consumption Prediction

Problem Type: Regression
Dataset: Kaggle – Water Usage Data
Objective: Predict water consumption.
Input Features: Household usage
ML Techniques: Random Forest
Metrics: RMSE
Outcome: Resource management.

93. Lifestyle-Based Disease Risk Prediction

Problem Type: Classification
Dataset: Kaggle – Lifestyle Dataset
Objective: Predict disease risk from habits.
Input Features: Diet, exercise
ML Techniques: Logistic Regression
Metrics: Accuracy
Outcome: Preventive healthcare.

94. Stock Volatility Prediction

Problem Type: Regression
Dataset: Kaggle – Stock Market
Objective: Predict stock volatility.
Input Features: Price history
ML Techniques: LSTM
Metrics: RMSE
Outcome: Financial risk analysis.

95. Online Gaming Player Retention Prediction

Problem Type: Classification
Dataset: Kaggle – Gaming Analytics
Objective: Predict player retention.
Input Features: Playtime, activity
ML Techniques: Random Forest
Metrics: Accuracy
Outcome: Game design optimization.

96. Traffic Signal Timing Optimization

Problem Type: Regression
Dataset: Kaggle – Traffic Signal Data
Objective: Predict optimal signal timing.
Input Features: Traffic flow
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Smart traffic systems.

97. Hospital Readmission Prediction

Problem Type: Classification
Dataset: Kaggle – Hospital Records
Objective: Predict patient readmission.
Input Features: Treatment history
ML Techniques: Logistic Regression
Metrics: Recall
Outcome: Healthcare quality improvement.

98. Climate Change Impact Prediction

Problem Type: Regression
Dataset: Kaggle – Climate Data
Objective: Predict climate impact indicators.
Input Features: Temperature, CO₂
ML Techniques: Linear Regression
Metrics: RMSE
Outcome: Environmental studies.

99. Job Salary Range Classification

Problem Type: Classification
Dataset: Kaggle – Job Postings
Objective: Classify jobs into salary ranges.
Input Features: Job description
ML Techniques: Naive Bayes
Metrics: Accuracy
Outcome: Employment analytics.

100. Customer Feedback Emotion Detection

Problem Type: NLP Classification

Dataset: Kaggle – Feedback Dataset

Objective: Detect emotions in feedback text.

Input Features: Text data

ML Techniques: TF-IDF, SVM

Metrics: Accuracy

Outcome: Customer sentiment insights.